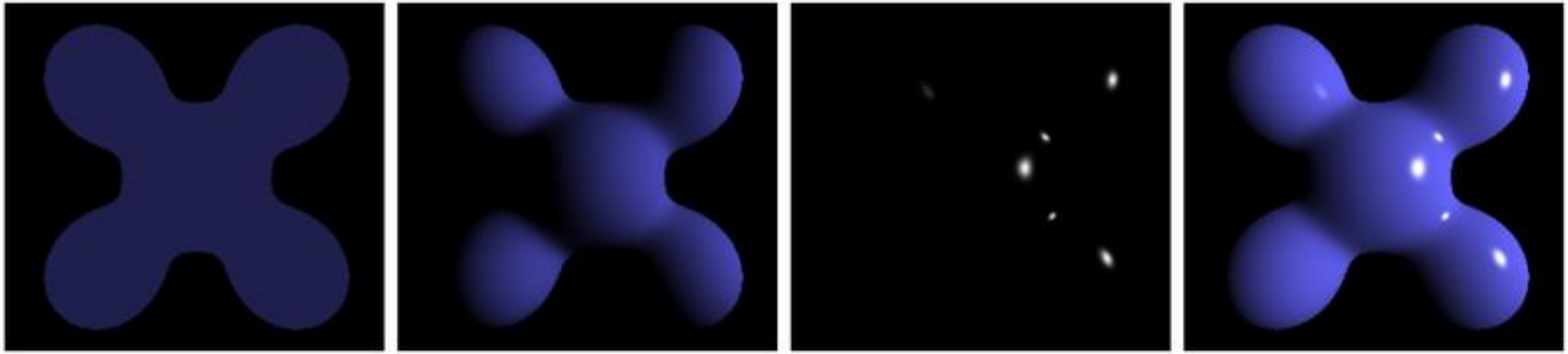
Componente ambiental, difusa y especular
Modelo completo de Phong



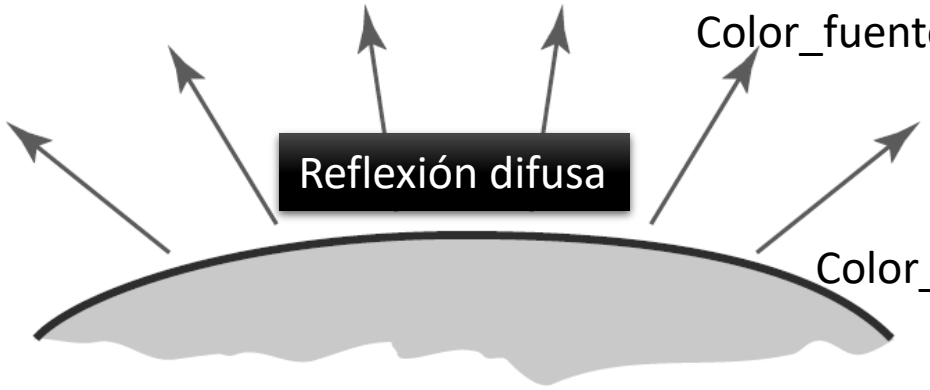
$$L_o = \left[k_a + k_d \left(\boldsymbol{n} \cdot \boldsymbol{l} \right) + k_s \left(\boldsymbol{v} \cdot \boldsymbol{r} \right)^q \right] \frac{L_i}{r^2}$$

			Propiedades materiales del objeto (color)			
			Ambiental	**Difusa**	**Especular**	
			$\mathrm{k}_a^o(\mathbf{R},\mathbf{G},\mathbf{B})$	$\mathbf{k}_d^o(\mathbf{R},\mathbf{G},\mathbf{B})$	$\mathbf{k}_s^o(\mathbf{R},\mathbf{G},\mathbf{B})$	
Propiedades de atenuación de color de la fuente de iluminación	Ambiental	$\mathbf{k}_a^L(\mathbf{R},\mathbf{G},\mathbf{B})$	$\mathbf{k}_a^o\mathbf{k}_a^L$			$\boldsymbol{k_a}$
	Difusa	$\mathbf{k}_d^L(\mathbf{R},\mathbf{G},\mathbf{B})$		$\mathbf{k}_d^o K_d^L$		$\boldsymbol{k_d}$
	Especular	$\mathbf{k}_s^L(\mathbf{R},\mathbf{G},\mathbf{B})$			$\mathbf{k}_s^o K_s^L$	$\boldsymbol{k_s}$

$$K_a = k_a \qquad K_d = k_d(n \cdot l) \qquad K_s = k_s(v \cdot r)^q$$

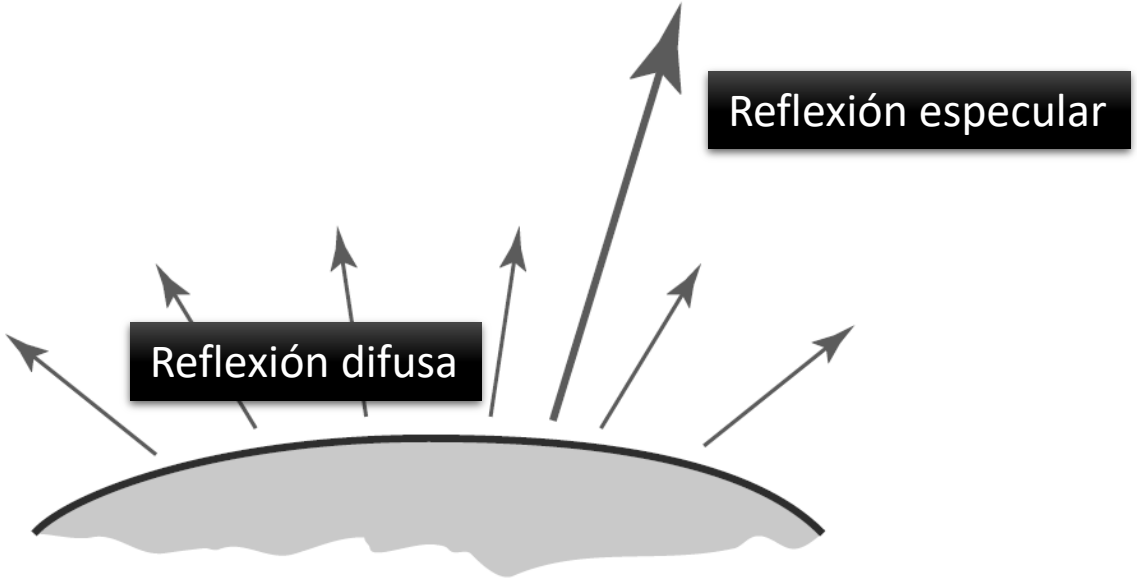
Reflexión superficial

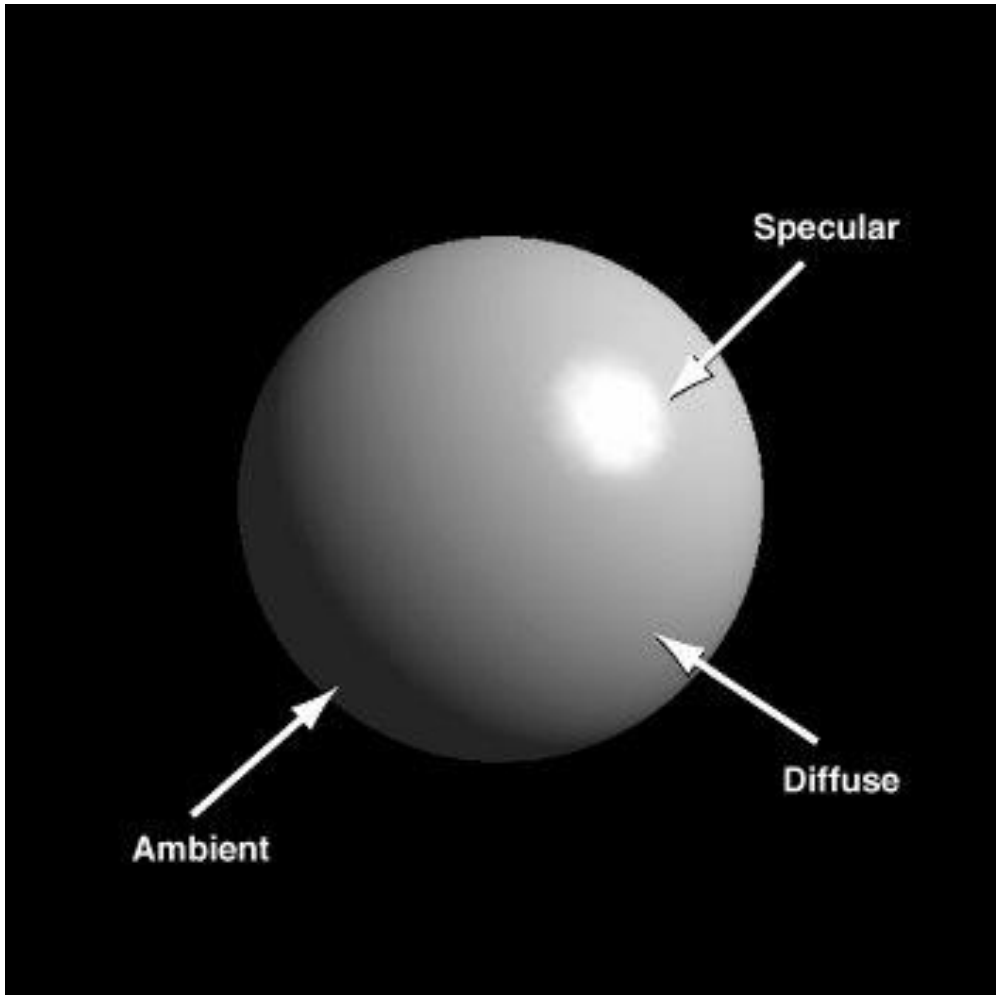
Color_fuente_luz_dif={0,1,0,1}



Color_objeto={1,1,1,1}

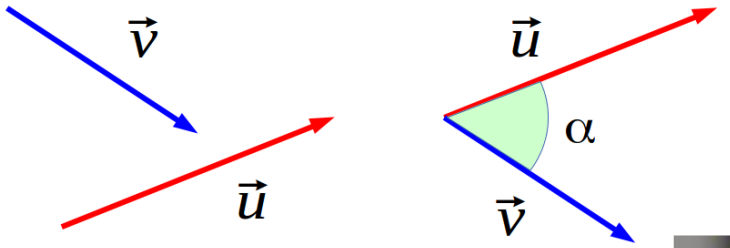
Color_final={0,0,1,1}





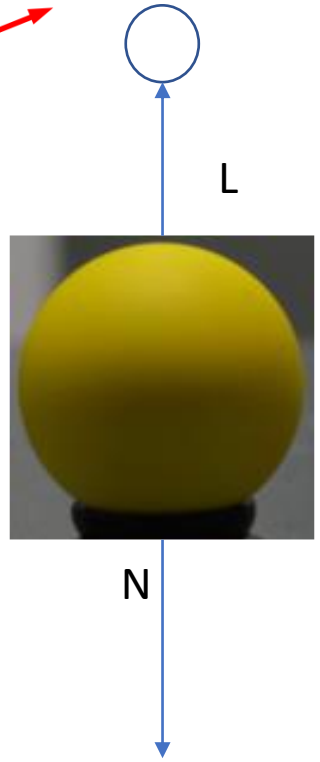
Reflexión difusa

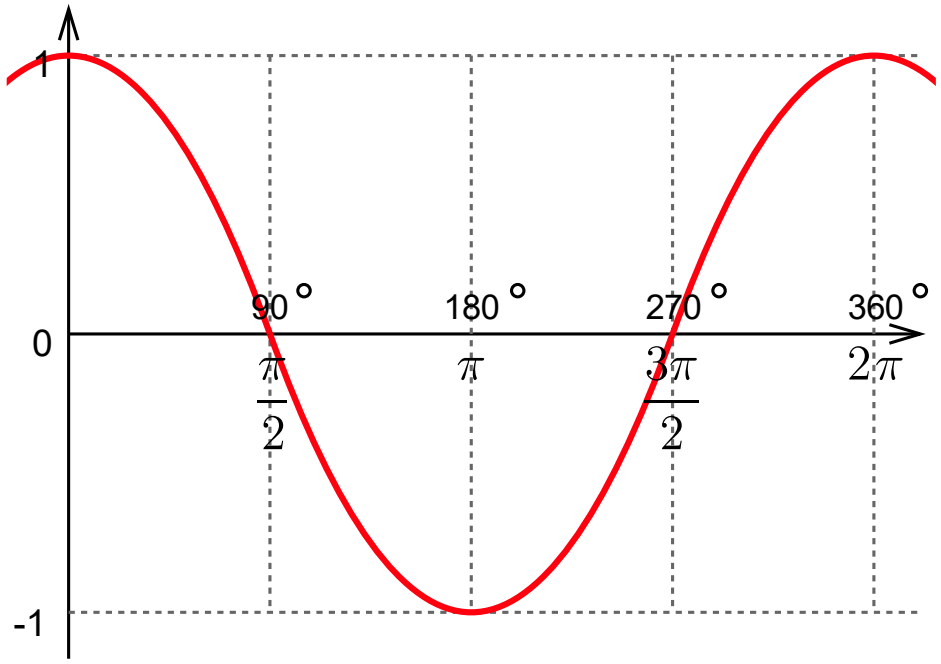
Producto escalar de dos vectores

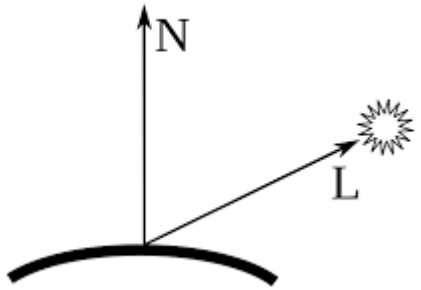


$$\vec{u}\cdot\vec{v}=|\vec{u}|\cdot|\vec{v}|\cdot\cos\alpha$$

$$\vec{u}\cdot\vec{v}=u_1\cdot v_1+u_2\cdot v_2$$

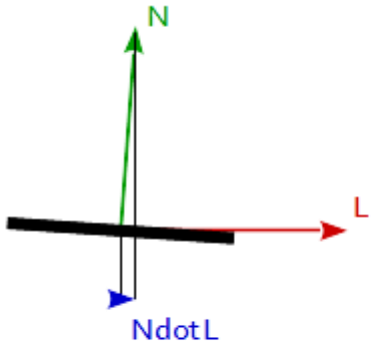


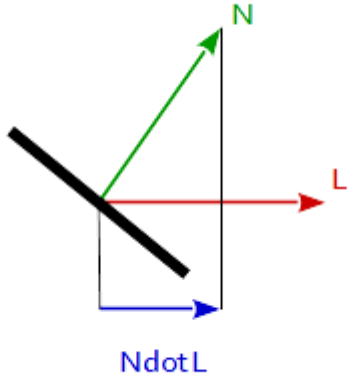


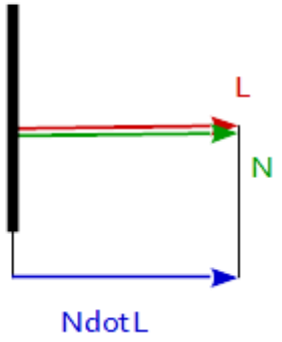


Dot = |N||L|cos(90) = 0

R,G,B



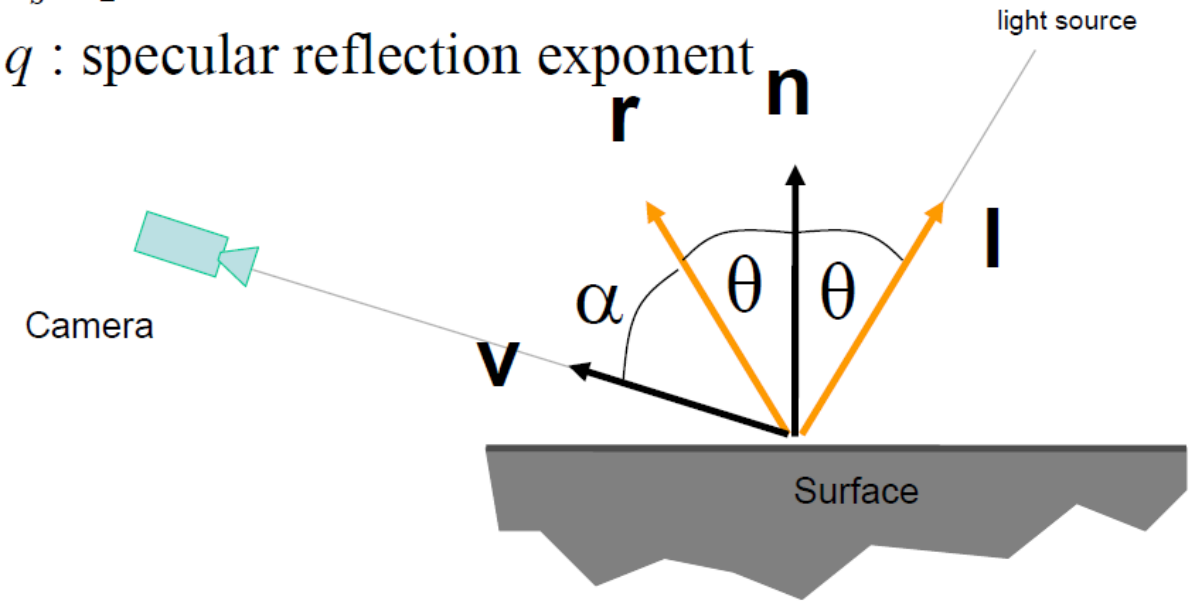


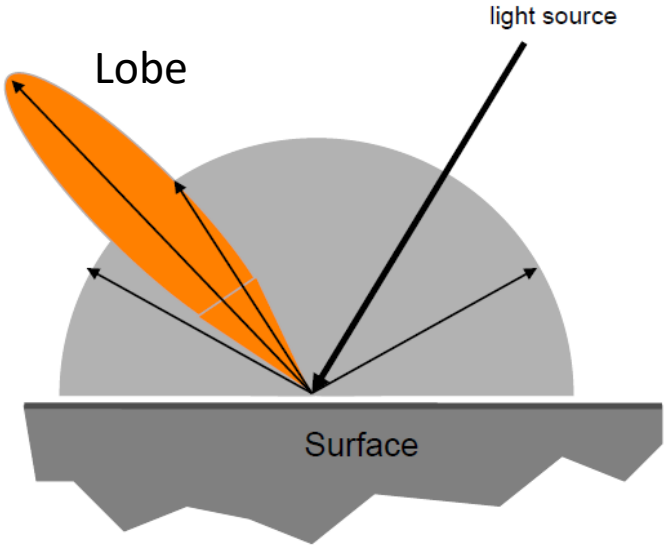


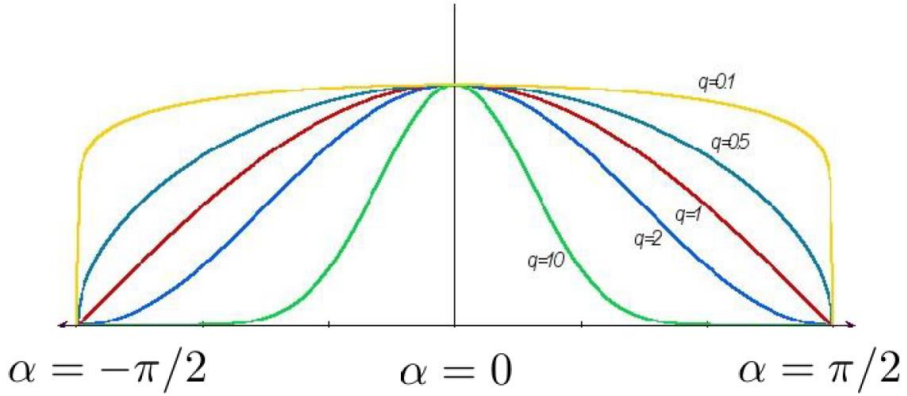
Modelo Especular de Phong

$$L_o = k_s (\cos\alpha)^q \frac{L_i}{r^2} = k_s (\boldsymbol{v} \cdot \boldsymbol{r})^q \frac{L_i}{r^2}$$

- Parameters
  - k_s: specular reflection coefficient
  - q : specular reflection exponent







Múltiples fuentes de iluminación

$$L = \sum_{i=1}^{N} \left[K_a^i + K_d^i + K_s^i \right] \frac{L_i}{r_i^2}$$